Lecture 25: ARCH Models

1 Dealing with nonstationarity of a time series

A conventional approach to handle unit-root nonstationarity is to use *differencing*. Although p_t is nonstationary, define the time series

$$r_t := p_t - p_{t-1} = \mu + a_t,$$

which is *stationary*. The time series $\{r_t\}_{t\geq 0}$ is the *first differenced* series of $\{p_t\}_{t\geq 0}$, i.e. $r_t := \Delta p_t$, where $\Delta := 1 - B$ is the difference operator. For a time series $\{p_t\}_{t\geq 0}$, it is possible that its first difference series Δp_t still has a unit root. This typically happens if the original time series p_t contain multiple unit roots. One may then consider its second, or d-th in general, differenced series $\Delta^d p_t$, until we obtain a weakly stationary time series.

An $ARMA(p, q)$ time series for $\Delta^d r_t$ is called an *autoregressive integrated moving average*, $ARIMA(p, d, q)$, for the time series r_t.

2 Trend and Seasonality

In general, a time series $\{r_t\}$ can be decomposed into "trend + seasonality + stationary time series". Indeed some financial time series, such as the quarterly earnings per share of a company, exhibits cyclical behavior. Both trends and seasonality can be removed by taking differences. For instance, suppose

$$r_t = f_t + y_t,$$

where $\{y_t\}_{t\geq 0}$, is a stationary process, and $f_t = at$ is a linear function representing a linear trend. Then, we have

$$\Delta r_t := r_t - r_{t-1} = \Delta f_t + \Delta y_t = a + \Delta y_t,$$

which is a stationary time series. Hence, the trend can be removed by taking differences. Next, suppose

$$r_t = g_t + y_t,$$

where $\{y_t\}_{t\geq 0}$ is a stationary process, and g_t is a period function of time t with periodicity s. Consider $\Delta_s g_t := (1 - B^s)g_t = 0$, where we recall that B is the back-shift operator. The operation $\Delta_s = (1 - B^s)$ is called seasonal differencing. Then

$$\Delta_s r_t = \Delta_s g_t + \Delta_s y_t = \Delta_s y_t$$

Hence, $\{\Delta_s r_t\}$ is stationary. Most economic data published by the US government are seasonally adjusted (e.g. the growth rate of gross domestic product, and the unemployment rate). Economic

growth is affected by different seasonal factors such as the weather and the holidays. Economists get a better picture of how an economy is moving when they adjust their analyses based on these factors. For example, roughly two-thirds of U.S. gross domestic product (GDP) is made up of consumer spending—which is a seasonal measure. The more consumers spend, the more the economy grows.

Remark 2.1. *How can we determine the ARMA signatures, i.e., whether we are dealing with an AR(p) model, or with a MA(q) model, or with an ARMA(p,q). We first plot the ACF and the PACF. Now, we know that for an AR(p) model, the ACF decays geometrically and PACF has p significant lags. If that is the case, then we have an ARMA(p,0) model, i.e., the signature is an AR model. Next, recall that a MA(q) model can be reproduced using an AR(∞) model. Hence, the ACF of a MA(q) model would have q significant lags (i.e., cut off at lag q), while the PACF would never cut off but decay geometrically. If our ARMA time series exhibit these characteristics, then we have a MA(q) model. If instead we have that both the ACF and PACF decay geometrically, then we have an ARMA model.*

3 ARCH Models and Stochastic Volatility

ARCH/GARCH models are suitable to model characteristics of volatility. Volatility is not directly observable. Consider, for example, the daily log returns of the IBM stock. The daily volatility is not directly observable from the return data because there is only one observation in a trading day. If intraday data such as 10-minute returns are available, then one could estimate the daily volatility even though the accuracy of such an estimate remains questionable.

Despite not being directly observable, the volatility presents characteristics that are commonly observed in asset returns. First, there exists volatility clusters, i.e., periods where volatility is high are followed by other periods with high volatility. Second, volatility evolves continuously over time, that is, volatility jumps are rare. Third, volatility does not diverge to infinity but varies within some fixed range. Statistically, this means that volatility is often stationary. Fourth, volatility reacts differently to a price increase or a price drop, also referred to as the *leverage* effect. Specifically, the observed tendency is for an asset's volatility to be negatively correlated with the asset's returns. Typically, rising asset prices are accompanied by declining volatility, and vice versa. The term "leverage" refers to one possible economic interpretation of this phenomenon, developed in Black (1976) and Christie (1982): as asset prices decline, companies become mechanically more leveraged because the relative value of their debt rises relative to that of their equity. As a result, it is natural to expect that their stock becomes riskier, hence more volatile. It has also been documented that the effect is generally asymmetric: other things equal, declines in stock prices are accompanied by larger increases in volatility than the decline in volatility that accompanies rising stock markets (see, e.g., Nelson, 1991; and Engle and Ng, 1993).